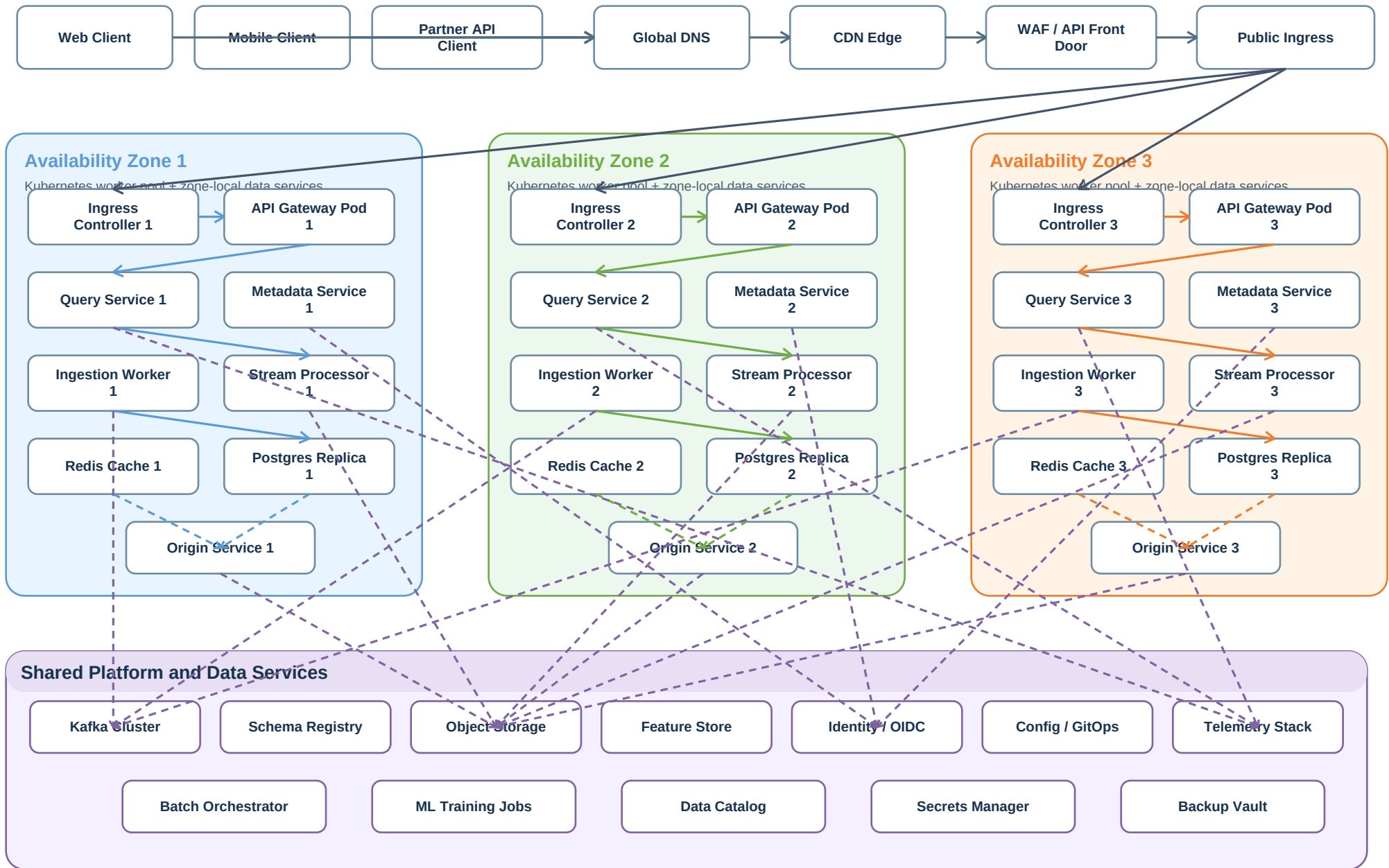


Synthetic Kubernetes Multi-AZ Data Platform - Architecture Stress Sample

Generic demo-safe regression architecture. Three clients, CDN, three availability zones, zone-local origins, Kubernetes workloads, shared platform services, and cross-AZ data paths.



Solid arrows = synchronous request / service flow. Dashed arrows = shared-service dependency or asynchronous data path.

Synthetic only: no real company names, logs, hosts, schemas, incidents, or proprietary topology.

Journey Map - Multi-AZ Kubernetes Data Platform

Designed to stress journey discovery, shared infrastructure, multi-rail understanding, stateful services, and conditional recovery paths.

1. Interactive Query

Web/Mobile/Partner Client -> DNS -> CDN -> WAF -> Public Ingress -> AZ Ingress -> API Gateway Pod -> Query Service -> Redis/Postgres -> response.

2. Origin Fetch / Delivery

CDN miss -> Public Ingress -> zone-local Origin Service -> Object Storage -> CDN Edge -> Client.

3. Streaming Ingestion

Partner API / producer -> Public Ingress -> Ingestion Worker -> Kafka -> Stream Processor -> Object Storage + Feature Store.

4. Metadata Lookup

Query Service -> Metadata Service -> Data Catalog -> Schema Registry -> Query Service.

5. Authentication / Authorization

Public Ingress or API Gateway Pod -> Identity / OIDC -> Secrets Manager -> request continues.

6. GitOps / Configuration

Config / GitOps -> Kubernetes workloads in all three AZs; supporting control-plane traffic.

7. Observability

All AZ workloads -> Telemetry Stack -> metrics, logs, and traces; remains separate from primary query rail.

8. Batch / ML

Batch Orchestrator -> Object Storage -> ML Training Jobs -> Feature Store -> Data Catalog.

9. Cross-AZ Resilience

Ingress can route to any healthy AZ. Postgres replicas, Kafka partitions, origins, and object data are replicated; candidate-only operational claims.

Component Glossary and Expected Generic Roles

No vendor-specific assumptions. Expected roles are regression hints only; deterministic evidence remains authoritative.

Component Pattern	Expected Role	Why it exists
Global DNS	entry / control	Provides global address resolution and route selection.
CDN Edge	entry / delivery	Caches assets and forwards misses toward an origin.
WAF / API Front Door	validation / control	Applies request filtering and policy before cluster ingress.
Public Ingress	entry / routing	Routes traffic to a healthy Kubernetes availability zone.
Ingress Controller	entry / control	Terminates cluster ingress and selects an internal service.
API Gateway Pod	entry / control	Provides API policy, request shaping, and service routing.
Query Service	processing	Serves interactive analytics and data access requests.
Metadata Service	processing / control	Resolves schemas, datasets, and catalog metadata.
Ingestion Worker	processing	Accepts incoming records and prepares them for streaming.
Stream Processor	processing	Transforms and enriches event streams.
Redis Cache	state / performance	Stores hot query state near application workloads.
Postgres Replica	state	Provides zone-local relational state and replica topology.
Origin Service	delivery / state access	Serves origin fetches and reads durable data from object storage.
Kafka Cluster	shared messaging	Carries asynchronous events across workloads and zones.
Object Storage	shared state / origin	Stores durable datasets, model artifacts, and origin objects.
Feature Store	shared state	Stores reusable features for analytics and ML.
Identity / OIDC	auth / validation	Provides shared authentication and token validation.
Config / GitOps	control	Distributes desired configuration to workloads.
Telemetry Stack	observability	Collects metrics, logs, and traces from every zone.

Expected Regression Targets for Cachey Notebook

Use this synthetic PDF to validate generalization after BUG-17F enterprise-topology hardening.

Expected Deployment Units

- Availability Zone 1
- Availability Zone 2
- Availability Zone 3

Expected Zone-Local Components

- Ingress Controller 1/2/3
- API Gateway Pod 1/2/3
- Query Service 1/2/3
- Redis Cache 1/2/3
- Postgres Replica 1/2/3
- Origin Service 1/2/3

Expected Shared Infrastructure Candidates

- Kafka Cluster
- Schema Registry
- Object Storage
- Feature Store
- Identity / OIDC
- Config / GitOps
- Telemetry Stack
- Data Catalog
- Secrets Manager
- Backup Vault

Expected Pattern Candidates

- multi-deployment or multi-AZ topology
- partial or mirrored workload replicas
- shared_infrastructure_topology

Demo safety rules

All names and relationships are synthetic, generic, and created only for regression testing.

No real company, partner, internal service, infrastructure identifier, log, schema, metric, incident, or customer data is included.

- request_journey
- content/data delivery journey
- state_journey